at the vessel base to a height of ~30 cm. The system was left at rest and temperature transients at the vessel walls and liquid contents were recorded with thermocouples. Due to space limitations only the essence of the results as summarized in Fig. 3 may be discussed here. The experimental data clearly show the downward development of the thermal boundary layer and they are in quantitative agreement with the $(\alpha_1 t)^{1/2}$ dependence. The spatial dependence of these profiles also agrees with that predicted for transient conduction into a semi-infinite domain with a constant temperature of the surface. Hence, this surface temperature may be estimated and was found to be in good agreement with our theoretical predictions as shown in Fig. 3. A rapid liquid surface temperature increase of 30–40 °C similar both in trend as well as in magnitude to that calculated for the Seveso reactor (Fig. 1) is thus deduced demonstrating the nature of the stratification phenomena and the simulation capability of our experimental setup.

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1. Hay, A. *Disasters* (to be published).
2. Hay, A. *Nature* **262,** 636 (1976); **267,** 384 (1977); **273,** 583 (1978).
3. Milnes, M. H. *Nature* **232,** 395 (1971).
4. Salomon, Ch. M. *Bericht uber den Störfall in Seveso, Sonderkolloquium am* 4 (Dechema-Haus, Frankfurt am Main, 1980).
5. Salomon, Ch. M. *Meet. Industrial Health and Safety Group of Soc. Chem. Ind.* London (1981).
6. Merzalanov, A. G. & Arerson, A. E. *Combust. Flame* **16,** 89 (1971).
7. Theofanous, T. G. *Meet. Industrial Health and Safety Group of Soc. Chem. Ind.* London (1981).
8. Theofanous, T. G. *Bericht uber den Störfall in Seveso, Sonderkolloquium am* 4 (Dechema-Haus, Frankfurt am Main, 1980).

Proterozoic magnetic overprinting of Archaean rocks in the Canadian Superior Province

Larry D. Schutts & David J. Dunlop

Department of Geology and Geophysics Division, Department of Physics, University of Toronto, Toronto, Canada M5S 1A7

We have measured the various components of natural remanent magnetization (NRM) exhibited by mafic extrusive and intrusive rocks from the Abitibi Subprovince of the Canadian Precambrian Shield. In addition to providing what seems to be a primary Archaean pole for the region, the results testify to three episodes of magnetic 'overprinting' at approximately 2,600, 2,100 and 1,800 Myr, respectively. The two earlier episodes coincide with the intrusion of Matachewan and Abitibi diabase dyke swarms, here and elsewhere in the Superior Province. Magnetic overprinting, in these cases, extends far beyond areas of surficial dyke exposure. The third episode postdates magmatic activity in the Abitibi region, but coincides in age with the 1,900–1,700 Myr Hudsonian orogeny in the Churchill and Bear Provinces and the Penokean orogeny in the Southern Province. Although magnetic overprints presumed to be of this age were known previously from Abitibi dykes[1] and Nipissing sills[2–4], our data are the first evidence of a 1,800 Myr magnetic overprinting on a regional scale in the Superior Province.

Palaeomagnetism is a powerful method of detecting and dating regional geological events in metamorphic terrains, particularly in Precambrian regions where complex structures, deep erosion, and various generations of burial and intrusion often confuse relations. Each geological event can result in thermal and/or chemical resetting of part of the pre-existing NRM. In favourable circumstances, careful multicomponent palaeomagnetic analysis can isolate vectorially superimposed NRM components corresponding to primary magnetization and one or more generations of NRM overprinting[5–8]. The geological events responsible are dated by comparing the palaeomagnetic pole corresponding to each NRM component with an established apparent polar wander path (APWP) for the continent or continental block.

The Abitibi greenstone belt of the Superior Province is particularly suited to a regional multicomponent palaeomagnetic study. The late Archaean and early Proterozoic units have been thoroughly mapped and the Laurentian APWP for this time interval is reasonably well established (Tracks 4, 5 and 6 of refs 9, 10). The Abitibi belt is characterized by mafic to felsic volcanic piles with coeval intrusions, volcanogenic sediments and intruding granitic batholiths[11,12]. In the Kirkland Lake, Ontario area, two such Archaean piles, plus the younger portion of a third pile, are preserved in a large east-plunging synclinorium[13]. The gabbroic intrusive bodies occur as both conformable sills[14] and as stocks, which seem to have been feeders for some of the flows[15]. Granitic bodies, apparently representing a later Archaean event, intrude the mafics on the north, west and south as batholiths and within the synclinorium as occasional stocks and plugs. Magmatic activity continued with intrusion of the north-south trending Matachewan diabase dyke swarm and essentially concluded with intrusion of the north-east trending Abitibi diabase dykes.

Archaean activity in the area was evidently episodic in occurrence. The youngest volcanic pile, which represents more than 70% of the 50,000 m total pile thickness[13], is bracketed in age by 2,710 ± 2 Myr and 2,703 ± 2 Myr U–Pb zircon dates[16]. These figures compare with 2,725 ± 2 Myr and 2,703 ± 3 Myr zircon ages bracketing the correlative sequence near Timmins, some 120 km to the west[17,18]. Granite ages are less narrowly defined. The oldest dates, determined by both K/Ar and Rb/Sr methods, fall within the 2,500–2,600 Myr range for the Round Lake Batholith[19,20]. The granites are cut by Matachewan dykes which, with a $\lambda(^{87}\mathrm{Rb})$ of 1.42×10^{-11} yr^{-1} (ref. 21), yield Rb/Sr isochron data implying an age of 2,633 ± 93 Myr (ref. 22). The relative age relation, however, is clear. The Matachewan dykes are seen to intrude the granite in many locations and absolute ages for both units probably fall within the 2,600–2,700 Myr span. The time of Abitibi dyke intrusion is more precisely defined and implies a discrete early Proterozoic event at 2,150 Myr (refs 22, 23).

For our study, 157 block samples were taken over an areal extent of 5,200 km^2 in the Kirkland Lake Matheson area. The suite of samples includes basalts and gabbros from the youngest synclinorium sequence, as well as diabases from 15 Matachewan and three Abitibi dykes. Several 2.5-cm cylindrical specimens were cut from each sample and subjected to detailed stepwise thermal (to 700 °C) and alternating field (to 100mT or 1,000 Oe) demagnetization, using standard commercial demagnetizers. NRM intensities and directions were measured on a computerized commercial spinner magnetometer, which provided a minumum detectable signal of 10^{-4} A m^{-1} (10^{-7} e.m.u. cm^{-3}) and on-line statistical analysis of results. (A more detailed analysis of the alternating field and thermal demagnetization results, and the results of experiments in progress designed to characterize more fully the magnetic remanence carriers, will be published elsewhere[24].)

Except for a few basalt and gabbro samples, in which an intermediate (~350 °C) Curie temperature carrier of magnetization was indicated, all samples exhibited maximum thermal demagnetization blocking temperatures near 580 °C, the Curie temperature of pure (titanium-free) magnetite. Basalt, gabbro and Abitibi dyke blocking temperatures tended to be distributed from 100 or 200 °C to 580 °C, while Matachewan dyke blocking temperatures were generally concentrated just below 580 °C. This evidence of relatively pristine magnetite as the principal NRM carrier is in accord with petrographical observations from this and other studies[15,25] implying only very low-grade regional metamorphism in the area.

Different methods were used to analyse directional changes during demagnetization. Stable endpoints, Briden Index criteria[26], Zijderveld-type plots[27], vector subtraction diagrams[28]

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Table 1 Summary of site mean palaeomagnetic results

	D	I	α_{95}	k	R	N	Palaeopole	
Characteristic directions								
Archaean basalt-gabbro	114°	−32°	11°	15	12.2	13	29° S	4° E
Matachewan dykes and baked contacts	21°	+13°	6°	44	13.7	14	45° N	70° E
Abitibi dykes and baked contacts	277°	+70°	8°	52	7.9	8	41° N	230° E
Secondary directions								
Archaean basalt-gabbro Matachewan-like	28°	+25°	11°	19	9.5	10	48° N	57° E
Archaean basalt-gabbro Abitibi-like	298°	+77°	9°	30	10.7	11	54° N	241° E
All units (sample averaged) post-Abitibi	166°	+56°	8°	14	26.1	28	4° S	291° E

D, I are declination and inclination of magnetization; α_{95}, k and R are Fisher statistical parameters; N is the number of sites averaged.

and converging remagnetization circles[29] all provided data relevant to the segregation of magnetic components. The isolated components were classified as either characteristic or secondary remanences. We define the 'characteristic NRM' in a multicomponent sum to be that component possessing the higher coercivities and blocking temperatures. A 'secondary NRM' is that component of lower coercivity and blocking temperature which overprints a characteristic NRM. Because our results imply that the reset magnetizations are of thermal as opposed to chemical origin, this scheme seems justified.

A standard three-tier (specimen/sample/site) analysis of characteristic and secondary NRM directions was carried out, a site being defined as a single stock, dyke, sill or flow and its baked contacts. Site-mean average magnetization directions are listed in Table 1 and plotted in Fig. 1. All NRM directions listed in Table 1 exhibit both normal and reverse polarities. Palaeopoles corresponding to each magnetization direction are shown in Fig. 2 and compared with the Tracks 5 and 6 of the North American APWP from refs 9, 10. Archaean ages have been updated to agree with the recent radiometric dates for the volcanics.

Abitibi and Matachewan dyke characteristic directions are well-defined and closely agree with the results of previous studies[1,10,30,31]. Evidence that these directions record the geomagnetic field direction at the time of dyke intrusion is provided by detailed Abitibi–Matachewan and Matachewan–gabbro contact tests in Munro Township. The pattern is the same in either case. Near the contact, NRM directions in the older unit have been reset parallel to those of the intrusive body. Well away from the contact, NRM directions in the older unit approach the characteristic direction. In an intermediate 'hybrid' zone, both NRMs are superimposed, the resultant direction swinging towards the older characteristic during magnetic cleaning. The maximum blocking temperature range of the overprinted NRM decreases in a regular fashion with distance from the contact. These data are strong evidence for partial to complete thermal resetting of primary NRM in the country rock at the time of dyke intrusion.

Basalt and gabbro samples were taken from the upper volcanic–plutonic pile of the synclinorium. The suite contains blocks from the Kinojevis tholeiite Group and the underlying Stoughton–Roquemaure and Larder Lake komatiite Groups. Three separate magnetic components, common to rocks from all three units, were isolated. Two of these components have the Abitibi and Matachewan dyke characteristic directions (Fig. 1*b*) and we believe them to be regional thermal overprints. The third component is similar to one found in other Superior Province Archaean units[10,32] and possibly records the geomagnetic field at the time these units were emplaced.

Several lines of evidence imply that the Abitibi-like and Matachewan-like components are secondary thermal overprints. They primarily affect the lower coercivity and blocking temperature spectra of the samples exhibiting them. They do persist throughout these spectra in some instances, but samples displaying this behaviour generally have coercivities and blocking temperatures centred about low absolute values. The directional changes with increasing demagnetization level are invariably in the sense suggested by the relative age data. Matachewan-like directions swing towards the older Archaean characteristic direction. Abitibi-like directions move towards the same Archaean direction or towards a Matachewan direction. These swings occur with both alternating field and thermal demagnetization. As both secondary and characteristic remanent magnetizations are lost by ~580 °C, the possibility of chemical overprinting seems remote. Partial thermal resetting of NRM as a result of, first, Matachewan and then Abitibi intrusion

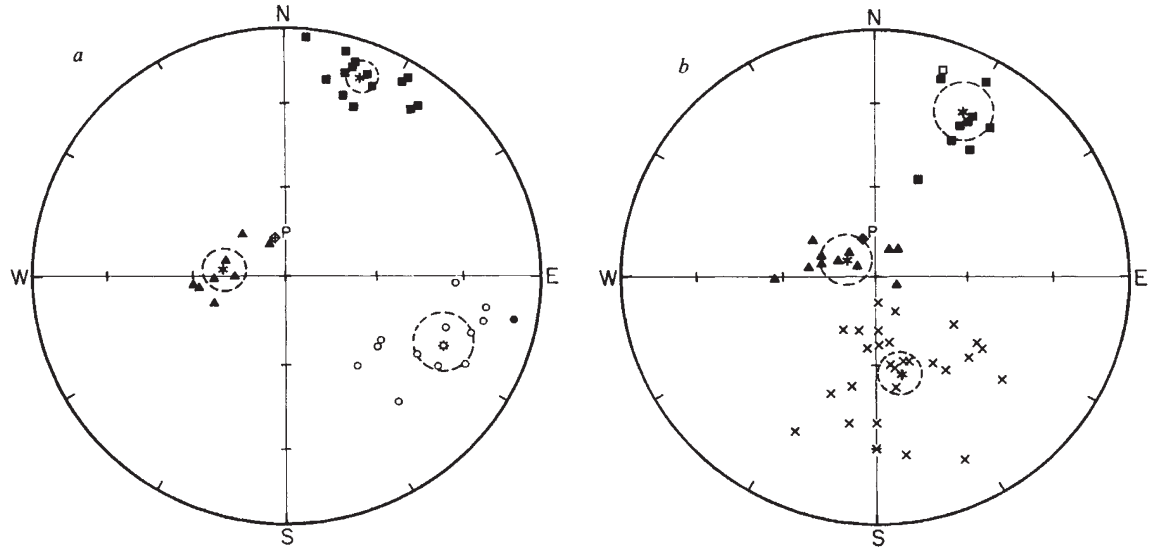


Fig. 1 Abitibi Subprovince site mean magnetization directions after optimum AF and/or thermal cleaning. Solid (open) symbols denote positive (negative) inclination. P (⊕) is the present Earth's field direction. ⁎, Average of site mean directions. *a*, Characteristic directions: ○, basalt–gabbro; ■, Matachewan diabase; ▲, Abitibi diabase. *b*, Secondary directions: ■, Matachewan-like; ▲, Abitibi-like; ×, Post-Abitibi (sample directions).

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seems to be the most likely explanation for the occurrence of both secondary components. As the overprints are generally found in areas containing little or no surficial expression of such dyke intrusion, the entire region was probably reheated, although the existence of dykes which do not break the present surface is also possible.

Averages of the basalt and the gabbro site-mean characteristic directions are nearly identical. Hence, results for both rock types are combined to give a 'basalt-gabbro' characteristic direction. Throughout the area sampled, the gabbros occur primarily as bodies intruding the highly folded basalts. The basalt directions, corrected for structure, become essentially random and a case might be made for regional resetting at the time of gabbro intrusion. The common characteristic direction, however, is also found in basalts far removed from any obvious intrusion. It is also found in a conformable gabbroic sill within the volcanics, which yields a negative fold test (McCool Township). These observations, combined with geological suggestions that the volcanics and plutonics are consanguineous[15] and that the structure in the volcanics developed during deposition[33], provide strong evidence for the primary nature of the 'basalt-gabbro' characteristic.

This scheme does not take into account the ubiquitous granite intrusions. They apparently represent an Archaean event which postdates synclinorium mafic activity, but predates Matachewan dyke intrusion. Current palaeomagnetic evidence does not rule out the basalt-gabbro characteristic direction being the result of a regional granitic intrusion heating. Low granite intrusion temperatures and gravity data suggesting that the larger bodies are limited to the synclinorium periphery[11,34] oppose this possibility. Ultimate resolution of the question, however, awaits detailed palaeomagnetic study of the granitic units.

Although plutonic and volcanic activity in the area essentially ceased with intrusion of the Abitibi dykes, a post-Abitibi, thermally-reset NRM component has been isolated in the lower coercivity–blocking temperature ranges of 28 of the samples studied. These components partially overprint each of the other characteristic and secondary directions already discussed and show no preference for rock type or geographic locality. They are, however, encountered only sporadically throughout the varous sites. As a result, sample means have been used to calculate the average direction and palaeopole.

Poles corresponding to these sample-mean directions are grouped around the 1,700–1,900 Myr segment of Tracks 4 and 5 (Fig. 2). Such components have previously been found in both Abitibi dyke[1] and Nipissing sill[2,3] samples from the region and have been assigned secondary magnetization status and late Aphebian age on the basis of palaeomagnetic work in the Cobalt area[4]. By using the $^{40}Ar/^{39}Ar$ step-heating variant of K/Ar dating, Hanes and York have uncovered evidence for a late Aphebian heating event in Munro Township[23]. Samples taken from an Abitibi dyke yielded a 2,150 Myr age for the dyke interior and 1,700–1,800 Myr ages for the edge zone contact with a Matachewan dyke. Several samples from this Matachewan dyke and one from the country rock gabbro also gave 1,700–1,800 Myr plateaus. The detailed Abitibi–Matachewan contact test made for this study was conducted where the same Abitibi dyke investigated by Hanes and York cut a Matachewan dyke 150 m east of the one they sampled. The post-Abitibi secondary direction was found in both units, most frequently near the contact. A late Aphebian age for this post-Abitibi NRM component is strongly implied.

More than half of the post-Abitibi directions discovered occurred in samples taken near intrusive contacts or zones of some faulting. A high heat flow in the region 1,700–1,900 Myr ago, facilitated by hot fluid transport through faults and intrusive contact zones of weakness, could explain both magnetic and argon resetting. A re-echoing of the Southern Province Penokean orogeny, for example, might be postulated as a source for the high heat flow. Although late Aphebian chemical resetting of remanence in some samples cannot be ruled out, Curie temperatures and preliminary microscopic examinations in

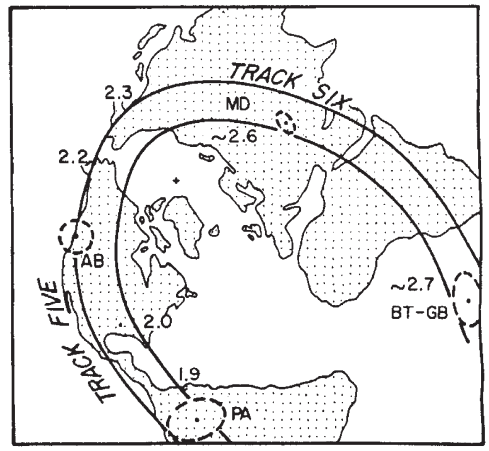


Fig. 2 Palaeomagnetic poles corresponding to Fig. 1 NRM directions as compared with the Laurentian APWP for the interval 1,900–2,700 Myr. BT–GB, basalt–gabbro characteristic. MD, Matachewan diabase characteristic—secondary. AB, Abitibi diabase characteristic—secondary. PA, post-Abitibi secondary.

reflected light suggest that thermal resetting should be the favoured hypothesis. Single Curie temperatures in the range 570–580 °C and intermediate deuteric alteration states with little sign of maghaemitization or haematization of the titanomagnetite phase imply no significant metasomatic alteration of the remanence carrier since formation.

Alternatively, the resetting may not be related to faulting and intrusive contacts on a regional scale and a late Aphebian unroofing event might be invoked to explain the younger ages and directions. The region was covered by Huronian sediments at ~2,300 Myr (ref. 35). A Schwarz-type[36,37] depth of burial calculation, using minimum blocking temperatures of the pre-overprint component, has been made for samples from the Munro Township Abitibi–Matachewan contact. The results imply an ambient temperature for the present erosional surface of 190 °C at the time of Abitibi dyke intrusion. Assuming a 25 °C km^{-1} geothermal gradient, this temperature implies a 7–8 km depth of burial at 2,150 Myr. The area was still covered, then, during the middle Aphebian, but a late Aphebian unroofing may certainly be postulated.

The close agreement of our independently dated post-Abitibi palaeopole for the Superior Province with palaeopoles ~1,800 Myr in age from the Churchill and Bear Provinces has broad implications for the tectonic history of Laurentia. This evidence implies that the Churchill orogen and the adjacent, relatively stable Superior craton occupied their present relative positions during the 1,900–1,700 Myr Hudsonian orogeny. The orogeny, therefore, may not have been the result of large-scale relative motions between these two blocks, such as are involved in modern day plate collision and subduction. The Hudsonian orogeny may represent an essentially ensialic reactivation of the Archaean Laurentian craton that incorporated both the Superior and Churchill Provinces.

Our preliminary Abitibi Subprovince results demonstrate that the geological history of complex Archaean terrains can be clarified with the help of palaeomagnetism. Resolved multi-component magnetizations can provide, in effect, a capsule history of thermal events throughout an entire region.

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1. Larochelle, A. *Can. J. Earth Sci.* **3**, 671 (1966).
2. Patel, J. P. & Palmer, H. C. *Can. J. Earth Sci.* **11**, 353 (1974).
3. Symons, D. T. A. & Londry, J. W. *Can. J. Earth Sci.* **12**, 940 (1975).
4. Roy, J. L. Lapointe, P. L. *Can. J. Earth Sci.* **13**, 749 (1976).
5. Buchan, K. L., Berger, G. W., McWilliams, M. O., York, D. & Dunlop, D. J. *J. Geomagn. Geoelectr.* **29**, 401 (1977).
6. McElhinny, M. W. & Senanayake, W. E. *J. geophys. Res.* **85**, 3523 (1980).
7. Beckmann, G. E. J. & Mitchell, J. G. *Earth planet. Sci. Lett.* **30**, 269 (1976).

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8. Morgan, G. E. *Nature* **259,** 382 (1976).
9. Irving, E. & McGlynn, J. C. *Trans. R. Soc.* **A280,** 433 (1976).
10. Irving, E. & Naldrett, A. J. *J. Geol.* **85,** 157 (1977).
11. Goodwin, A. M. & Ridler, R. H. *Geol. Surv. Canad. Pap.* 70–40, 1 (1970).
12. Goodwin, A. M. *Geol. Ass. Can. Spec. Pap.* **16,** 205 (1977).
13. Jensen, L. S. *Ont. Geol. Surv. Misc. Pap* No. 90, 64 (1979).
14. Pyke, D. R., Naldrett, A. J. & Eckstrand, O. R. *Bull. geol. Soc. Am.* **84,** 955 (1973).
15. Jensen, L. S. *Ont. Geol. Surv. Rep.* 165 (1978).
16. Nunes, P. D. & Jensen, L. S. *Ont. Geol. Surv. Misc. Pap.* 92 (1980).
17. Pyke, D. R. *Ont. Geol. Surv. Misc. Pap.* No. 82, 73 (1978).
18. Nunes, P. D. & Pyke, D. R. *Ont. Geol. Surv. Misc. Pap.* 92 (1980).
19. Aldrich, L. T. & Wetherill, G. W. *J. geophys. Res.* **65,** 337 (1960).
20. Purdy, J. W. & York, D. *Can. J. Earth Sci.* **5,** 699 (1968).
21. Steiger, R. H. & Jäger, E. *Earth planet. Sci. Lett.* **36,** 359 (1977).
22. Gates, T. M. & Hurley, P. M. *Can. J. Earth Sci.* **10,** 900 (1973).
23. Hanes, J. A. & York, D. *Can. J. Earth Sci.* **16,** 1060 (1979).
24. Schutts, L. D. (in preparation).
25. Jolly, W. T. *Can. Miner* **12,** 499 (1974).
26. Briden, J. C. *J. geophys. Res.* **77,** 1401 (1972).
27. Zijderveld, J. D. A. in *Methods of Paleomagnetism* (eds Collinson, D. S., Creer, K. M. & Runcorn, S. K.) 172 (Elsevier, Amsterdam, 1967).
28. Halls, H. C. *Earth planet. Sci. Lett.* **43,** 303 (1979).
29. Halls, H. C. *Phys. Earth planet. Inter.* **16,** 1 (1978).
30. Strangway, D. W. *J. Geol.* **72,** 648 (1964).
31. Fahrig, W. F., Gaucher, E. H. & Larochelle, A. *Can. J. Earth Sci.* **2,** 278 (1965).
32. Dunlop, D. J. *Phil. Trans. R. Soc.* (in the press).
33. Jensen, L. S. *Toronto '78 Field Trips Guidebook,* 237 (Geological Association of Canada, 1978).
34. Middleton, R. S. *Ont. Div. Mines, Geosci. Rep.* 135 (1976).
35. Fairbairn, H. W., Hurley, P. M., Card, K. D. & Knight, C. J. *Can. J. Earth Sci.* **6,** 489 (1969).
36. Schwarz, E. J. *Can. J. Earth Sci.* **14,** 82 (1977).
37. Buchan, K. L. & Schwarz, E. J. *Can. J. Earth Sci.* (submitted).

Low-angle normal faults in the Basin and Range Province: nappe tectonics in an extending orogen

Brian Wernicke

Department of Earth and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA

Cenozoic normal fault mosaics bounded beneath by a basal fault in the Basin and Range Province (BRP) have traditionally been described either in terms of large-scale surficial gravity sliding or by some form of *in situ* lower plate accommodation. I suggest here that these areas may be an extensional analogue to thin-skinned compressional orogens, a process which may even dominate active BRP tectonics.

These terranes consist of a mosaic of rock sheets bounded by high- and low-angle normal faults[1,2] which resemble (and have been misinterpreted as) compressional nappe piles. In many areas, the normal faults 'bottom out' into a single, subhorizontal fault zone below which either minor or no tectonism has occurred[3–6], or a roughly synchronous, more ductile style of deformation is present[7,8]. The minimum areal extent of the basal faults is typically measured in thousands of square kilometres. For example, in east-central Nevada, normal fault systems developed in Palaeozoic miogeoclinal strata, Mesozoic and Tertiary intrusives, and Tertiary volcanics end abruptly at a basal fault (Snake Range decollement[3], Fig. 1) accompanied by a thin (0–100 m) zone of tectonite marble. Offsets on the basal and higher faults are difficult to contrain, but Nelson reports a 20-km minimum horizontal offset for one fault alone in the Deep Creek Range[9]. My palinspastic reconstruction of the Northern Egan and Cherry Creek ranges (Fig. 2) indicates a 16.5-km offset for one sheet in the mosaic, and the per cent increase in original width of the area is ~300%. As the basal and higher faults in Fig. 1 are commonly subparallel to bedding and often cut out thousands of metres of section, total offsets are probably at least in the range of several tens of kilometres. Most authors except one[10] who have mapped in the area outlined in Fig. 1 agree that the transport direction of the allochthonous sheets was eastwards. The faults also tend to cut consistently downsection to the east[11,12], (Fig. 2). This terrane apparently has every attribute of a thin-skinned compressional orogen (basal decollement, far-travelled allochthons, consistent direction of section transgression and transport) except (1) the low-angle faults habitually omit rather than repeat section, (2) associated high-angle faults are predominantly normal instead of reverse, and (3) folding is of subdued importance[13]. Though early workers attempted to correlate these faults with Mesozoic thrusting in the Sevier belt to the east[3], there is considerable evidence to support a Tertiary, extension-related origin for most of the deformation[13–15].

Models for these areas must account for three observations: (1) the basal and higher faults predominantly represent extension; (2) the rocks below the basal faults are tectonically inert during transport of the allochthonous sheets and high-angle fault blocks; and (3) the areal extent and transport distances of the allochthons are large. Current models fall into two categories. The first is a localized gravity-slide model in which domal upwarps shed their cover in a megalandslide fashion[15–18] (Fig. 3*a*). The second invokes penetrative ductile stretching and/or igneous intrusion to drive extension in the immediately overlying brittle fault mosaic[2,7,8,19,20] (Fig. 3*b*). Landslide models are weakened by the scarcity of geometrically required areas of shortening correlative in age, size and direction of transport with the extensional terranes. Thus, the model agrees with observations (2) and (3), but not with observation (1). *In situ* ductile stretching or intrusion models lose appeal when it is noted that most metamorphic fabrics and igneous bodies (if any) beneath the basal faults are of inappropriate age or geometry to be rigorously linked to normal faulting[5,6,21,22]. Thus, the model is in harmony with observations (1) and (3), but not with observation (2).

The key to the problem may lie in our understanding of compressional orogens, in which the horizontal movement of thin sheets of rock above an underthrusting slab may occur many

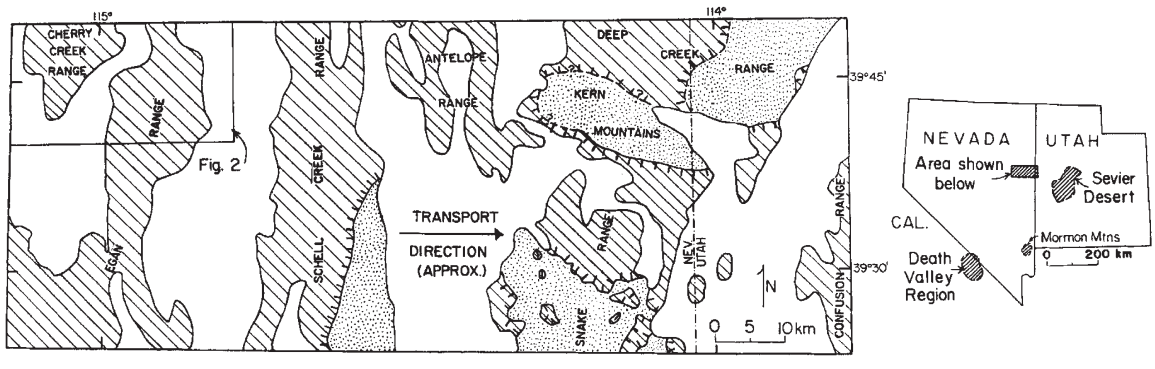


Fig. 1 Location map and tectonic interpretation of east-central Nevada, showing autochthon (stippled), basal fault zone (line with tick marks, queried where uncertain) and normal fault mosaic (hatched). Basal fault in the Schell Creek Range is locally tilted steeply westwards, implying continued deformation after it ceased moving, possibly on a lower basal fault. Tilting and faulting in the Northern Egan Range may be related to this second event rather than to movement on the basal fault shown here.

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